

Deep learning-derived orthogonal minimum joint space width improves radiographic assessment of knee osteoarthritis severity and progression

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Abstract

Purpose: To evaluate the performance of a novel orthogonal minimum joint space width, defined as the narrowest distance measured orthogonally to the tibial joint line within a standardised medial axis, for detecting and monitoring knee osteoarthritis progression. This artificial intelligence-derived metric, obtained through deep learning, was compared with conventional fixed-location joint space width measures.

Methods: A total of 15,313 knee radiographs from the Osteoarthritis Initiative (OAI), spanning baseline to 72-month follow-up, were analysed in this retrospective cohort study. Orthogonal minimum joint space width was automatically measured using a deep learning model. Discriminative ability for cross-sectional joint space narrowing severity was assessed using the area under the receiver operating characteristic curve. Longitudinal responsiveness was evaluated with the standardised response mean and relative standardised response mean, computed from pooled 12-month intervals and stratified by baseline narrowing grade. Variability was estimated through 1000 bootstrap resamples. Performance was compared with fixed-location measures at 22.5% and 25.0% from the medial edge of the tibial plateau.

Results: Orthogonal minimum joint space width demonstrated strong discriminative performance, with area under the curve values of 0.86, 0.95 and 0.97 for joint space narrowing greater than grades 0, 1 and 2, respectively – exceeding the corresponding values for fixed-location measures by absolute margins of 0.04–0.08. Pooled relative standardised response mean analysis showed consistently higher responsiveness for orthogonal minimum joint space width, with median values of 0.97, 0.95 and 0.91 for baseline narrowing grades 0, 1 and 2, representing approximately 2%–4% greater responsiveness than fixed-location measures across all strata.

Conclusions: Orthogonal minimum joint space width, automatically quantified by deep learning, enhances both sensitivity and specificity for assessing knee osteoarthritis severity and progression. This metric may

Abbreviations: CNN, convolutional neural network; DEKR, deformable keypoint regression; JSN, joint space narrowing; JSW, joint space width; JSW225, joint space width at 22.5% of tibial plateau width; JSW250, joint space width at 25.0% of tibial plateau width; KLG, Kellgren–Lawrence grade; MRI, magnetic resonance imaging; OA, osteoarthritis; OAI, osteoarthritis initiative; oJSW, orthogonal minimum joint space width; ROC, receiver operating characteristic; rSRM, relative standardised response mean; SRM, standardised response mean.

serve as a robust and reproducible structural imaging endpoint for clinical trials and longitudinal research.

Level of Evidence: Level II.

KEYWORDS

biomarkers, deep learning, joint space width, knee osteoarthritis, radiography

INTRODUCTION

An estimated 528 million individuals are living with osteoarthritis (OA) [25] (with the knee being the most frequently affected joint) – and is a leading cause of pain and disability in older adults [13]. With aging populations and rising obesity, the burden of knee OA continues to increase [14], underscoring the need for more effective interventions and metrics to monitor disease progression.

Radiographic assessment of joint space width (JSW) remains a widely accepted surrogate marker for evaluating cartilage thickness and monitoring OA progression due to its accessibility, low cost and extensive validation in clinical research [4, 6]. Traditional JSW measurements include the minimum joint space width (mJSW), which captures the narrowest distance between joint line curves [9], and fixed-location JSW (fJSW) metrics such as JSW225 and JSW250, defined at 22.5% and 25.0% from the medial edge of the femoral condyle [8, 18]. While mJSW reflects the point of maximal cartilage loss, it is limited by poor responsiveness in advanced stages of knee OA [20]. Conversely, fixed-location measurements have demonstrated greater longitudinal sensitivity [18], particularly in higher Kellgren–Lawrence (KL) grades, but may fail to capture individual-specific anatomical variability and may miss the true site of maximal narrowing.

Recent advances in artificial intelligence (AI), particularly deep learning techniques, have enabled fully automated and reproducible assessments of JSW [15, 17]. These models reduce inter-reader variability and enhance detection of structural changes, especially when applied to standardised anatomical landmarks. However, the suggested parameters have not been validated in a longitudinal dataset for its responsiveness nor have been compared with the previous measurable JSW parameters such as mJSW or fJSW.

To overcome these limitations, the authors devised a novel AI-derived metric – *orthogonal minimum joint space width (oJSW)* – measured using a convolutional neural network (CNN) that automatically identifies the narrowest point within a predefined medial tibiofemoral region. This AI-based approach provides automated, and reproducible measurements, minimising observer

variability and geometric inconsistencies present in manual or fixed-location methods.

The primary objective of this study was to evaluate the discriminative performance and longitudinal responsiveness of oJSW in comparison to established fJSW metrics (JSW225 and JSW250), using data from the Osteoarthritis Initiative (OAI) [19]. It was hypothesised that oJSW would demonstrate superior sensitivity in distinguishing JSN severity and greater responsiveness to structural progression over time.

METHODS

Study design and dataset

This retrospective cohort analysis utilised data from the OAI, comprising 15,313 fixed-flexion posteroanterior knee radiographs from 3,855 participants obtained from baseline up to the 72-month follow-up visits. Radiographs were included if corresponding medial joint space narrowing grades and metadata (age, sex, body mass index [BMI] and Kellgren–Lawrence grade [KLG]) were available. Images with incomplete data or inadequate quality were excluded. Each participant contributed one or both knees across multiple time points, allowing longitudinal assessment of structural change.

This study was conducted using publicly available data from the OAI, which received institutional review board approval at all participating centres. As only de-identified data were used, additional ethical approval and informed consent were not required for the present analysis.

JSW measurement and structural outcomes

The primary imaging marker, oJSW, was measured using a deep learning-based automated pipeline developed and validated in-house. This CNN-based system identifies the narrowest medial tibiofemoral joint space width along a fixed axis. The entire process is end-to-end and requires only input radiographs, without manual annotation or adjustment.

AI-based measurement pipeline

The pipeline comprises two major components: keypoint detection and anatomical joint line segmentation.

1. Keypoint Detection for Tibial Joint Boundaries: A modified DEKR (Deformable Keypoint Regression) model [10], initially designed for human pose estimation, was adapted and trained on annotated knee radiographs to detect the medial and lateral endpoints of the tibial joint line. These endpoints establish a normalised horizontal axis (0–1) across the tibial plateau.
2. Medial and Lateral Zone Definition: Using the normalised coordinate system, two fixed zones were defined:
 - Medial region: $x = 0.000–0.275$
 - Lateral region: $x = 0.725–1.000$
3. Joint Line Segmentation: Independent U-Net architectures [23] were trained to segment the femoral and tibial joint lines. The resulting segmentation outputs – *femur_joint_line* and *tibia_joint_line* – provided the vertical (y-axis) coordinates needed for JSW calculation.
4. oJSW Calculation: For each discrete x-coordinate within the defined medial and lateral zones:
 - The algorithm checked whether both femoral and tibial segmentation lines provided valid y-values at that x-position.
 - If valid, the vertical distance between the femur and tibia lines (i.e., $JSW = tibia_y - femur_y$) was computed.
 - The minimum JSW encountered across each zone was retained as the oJSW for that compartment (medial or lateral).
5. Knee Laterality Handling: To ensure accurate compartment classification, the system identified knee laterality from the DICOM metadata or filename heuristics. For left knees, the 0–0.275 region was classified as medial and 0.725–1 as lateral; the opposite convention was used for right knees.

This framework guarantees consistent anatomical interpretation across laterality and image orientation, preventing medial–lateral misclassification. Comparator fixed-location JSW values at 22.5% and 25.0% from the medial condylar edge (JSW225 and JSW250) were extracted from the OAI fixed-location dataset. Figure 1 illustrates the anatomical definition of each metric.

Structural outcome

Medial JSN was graded using the Osteoarthritis Research Society International (OARSI) atlas criteria [1]. For cross-sectional discriminative analysis, JSN was dichotomised at three thresholds: $JSN > 0$, $JSN > 1$ and $JSN > 2$. Longitudinal progression was assessed by calculating 12-month changes in JSW (ΔJSW) and corresponding changes in JSN grade (ΔJSN) across consecutive time intervals (0–12, 12–24, 24–36 and 36–48 months).

Statistical analysis

Discriminative performance was assessed using receiver operating characteristic (ROC) analysis, with the area under the curve (AUC) computed to evaluate the ability of each JSW metric to distinguish JSN severity thresholds ($JSN > 0$, > 1 , > 2). Pairwise comparisons of AUCs between metrics were conducted using the DeLong test. Longitudinal responsiveness was quantified using the standardised response mean (SRM), defined as the mean change in 12-month ΔJSW divided by its standard deviation. Relative responsiveness was evaluated using the relative SRM (rSRM), calculated as $SRM_1 / (SRM_1 + SRM_0)$ where SRM_1 is SRM for $\Delta JSN = 1$ and SRM_0 is SRM for $\Delta JSN = 0$. rSRMs were derived from ΔJSW values pooled across consecutive 12-month intervals (e.g.,

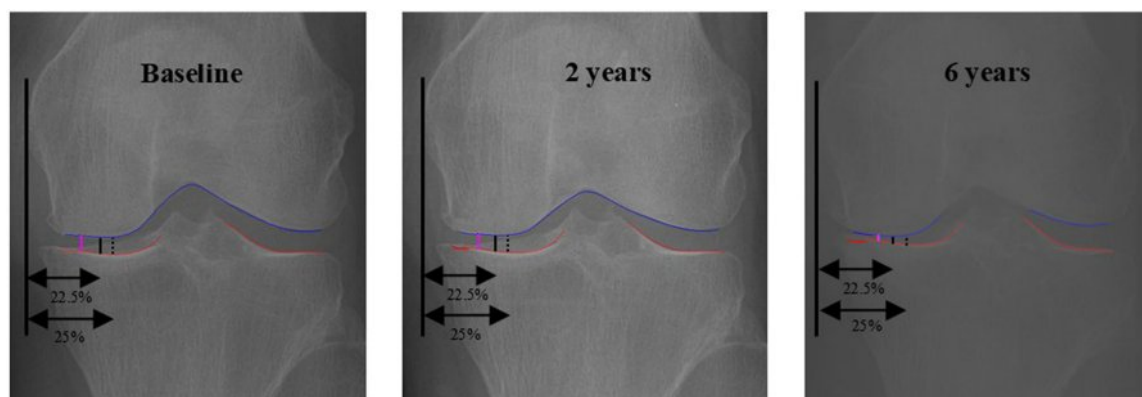


FIGURE 1 The blue line curve represents the femoral joint line and the red line curve represents the tibial joint line. The violet line represents the oJSW, the black solid line represents fJSW 225, and the dotted black line represents fJSW 250. From baseline (left) to 2- (middle) and 6-year (right) follow-up, oJSW demonstrates a reliable, robust, and reproducible structural imaging endpoint.

0–12, 12–24, 24–36, 36–48 months), but stratified by baseline JSN grade. Bootstrap resampling with 1000 iterations was used to estimate 95% confidence intervals and assess variability. All analyses were performed using Python (version 3.10).

No formal power calculation was performed, as this study utilised the entire publicly available OAI dataset. The large sample size ensured sufficient statistical power for detecting differences across metrics and subgroups.

RESULTS

The analysis included 15,313 knee radiographs from 3,855 unique participants, spanning baseline and follow-up visits. The mean age at baseline was 60.9 ± 11.0 years, and 57.9% of participants were female. The mean BMI was 28.7 ± 4.7 kg/m². The distribution of Kellgren–Lawrence (KLG) grades at baseline is shown in Table 1.

Discriminative performance

The oJSW demonstrated superior discriminative ability compared to JSW225 and JSW250 in differentiating medial JSN severity thresholds. For JSN > 0, the area under the ROC curve (AUC) for oJSW was 0.86, compared to 0.81 for JSW225 and 0.78 for JSW250. At higher thresholds, AUCs were 0.95 (oJSW), 0.91 (JSW225) and 0.88 (JSW250) for JSN > 1 and 0.97 (oJSW), 0.95 (JSW225) and 0.93 (JSW250) for JSN > 2. ROC curves consistently demonstrated stronger discriminative performance of oJSW across all thresholds (Figure 2). Pairwise DeLong tests confirmed the statistical

superiority of oJSW versus JSW225 (Δ AUC = 0.053, 0.043 and 0.016, respectively) and JSW250 (Δ AUC = 0.075, 0.068 and 0.031; all $p < 0.001$).

Quantile distributions of Δ JSW, pooled across 12-month intervals and stratified by baseline JSN grade and Δ JSN, further supported these findings. For oJSW, the median change declined stepwise with progression: from 0.00 (Δ JSN = 0) to −0.70 (Δ JSN = 1) and −1.90 (Δ JSN = 2). Compared to JSW225 and JSW250, oJSW showed the most consistent and pronounced gradient in both median and interquartile range (Figure 3).

Longitudinal responsiveness

Twelve-month standardised response means (SRMs) for Δ oJSW, Δ JSW225 and Δ JSW250, stratified by baseline JSN and Δ JSN, are presented in Figure 4. In knees without structural progression (Δ JSN = 0), SRMs were low and near zero for all metrics. Among knees with mild progression (Δ JSN = 1), oJSW showed strong responsiveness (SRM = −1.10), comparable to JSW225 (−1.49) and JSW250 (−1.35). In more advanced cases (Δ JSN = 2), all three metrics exhibited high SRMs, with JSW225 showing the largest value (SRM = −3.08) among knees with baseline JSN = 0.

Relative responsiveness, assessed using bootstrapped relative SRM (rSRM) from pooled 12-month intervals, consistently favoured oJSW. Stratified rSRMs for oJSW were 0.97, 0.95 and 0.91 for baseline JSN grades 0, 1 and 2, respectively, exceeding those of JSW225 (0.95, 0.92 and 0.89) and JSW250 (0.94, 0.92 and 0.89). oJSW showed numerically higher relative responsiveness than fixed-location metrics (Δ rSRM $\approx$ 0.02–0.03), but bootstrap comparisons did not reach statistical significance across strata (all $p > 0.05$). These results demonstrate that oJSW maintained the highest relative SRM across all JSN grades, indicating greater specificity to true structural progression compared to JSW225 and JSW250 (Figure 5).

DISCUSSION

Compared to standard fixed-location measurements (JSW225, JSW250), oJSW demonstrated superior discriminative validity for JSN grades and greater specificity to longitudinal structural changes. These results suggest that oJSW offers advantages as a responsive, patient-specific imaging endpoint in both clinical research and practice.

The superior performance of oJSW stems from its methodology. Traditional fixed-location metrics measure JSW at predefined points (typically 22.5% or 25% from the medial tibial edge) [18]. Although standardised and reproducible, this assumes uniform

TABLE 1 Demographic information of the individuals included in the study at baseline.

N = 3855	
Age (years)	60.9 (11.0)
Female (%)	57.9%
BMI (kg/m ²)	28.7 (4.7)
KLG	
0	880 (22.8%)
1	590 (15.3%)
2	1493 (38.7%)
3	743 (19.3%)
4	149 (3.9%)

Note: Numbers in parenthesis represents standard deviation for continuous variables and proportion percentages for categorical variables.

Abbreviations: BMI, body-mass index; KLG, Kellgren–Lawrence grade.

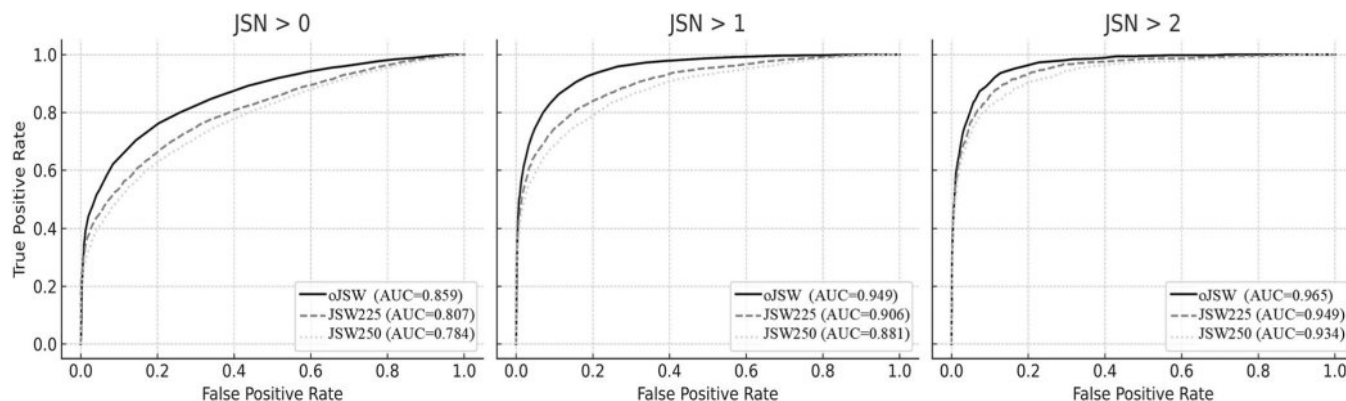


FIGURE 2 ROC curves comparing oJSW, JSW225 and JSW250 for detecting medial JSN severity. JSN > 0 (left), JSN > 1 (middle) and JSN > 2 (right). Curves are shown in grayscale; AUC values are inset.

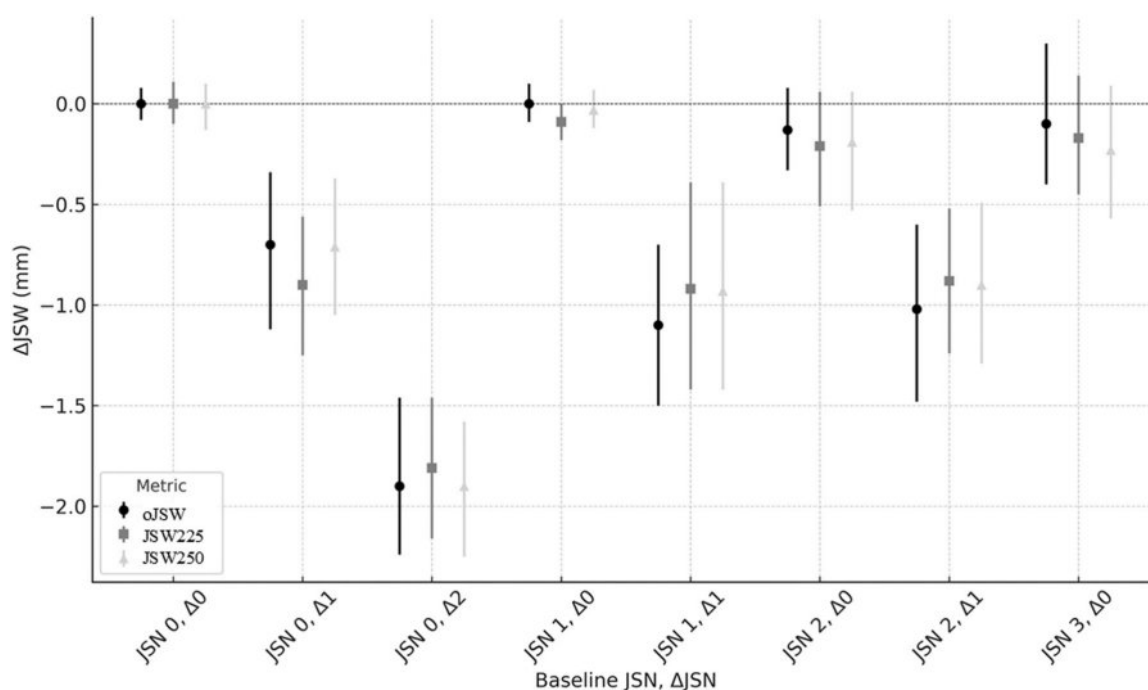


FIGURE 3 Bootstrapped mean 12-month changes in joint space width (Δ JSW) for oJSW, JSW225 and JSW250, stratified by baseline medial JSN grade and Δ JSN category. Points represent the mean Δ JSW, and error bars indicate 95% confidence intervals derived from 1000 bootstrap iterations. oJSW is shown as black circles, JSW225 as grey squares and JSW250 as light grey triangles.

disease distribution, which does not reflect the anatomical and pathological variability seen in clinical osteoarthritis populations [16]. Because OA often progresses asymmetrically – especially in varus or valgus knees – fixed-point measures may miss the true site of maximal cartilage loss, underestimating severity. In contrast, oJSW uses a deep learning-based approach to automatically identify the narrowest joint space within a defined medial region (0%–27.5% of the tibial plateau), combining the individualised sensitivity of mJSW with the standardisation of a fixed anatomical zone [18]. Unlike earlier mJSW techniques requiring manual or semi-

automated tracing prone to observer variability [9], the deep learning model provides a fully automated, reproducible solution applicable at scale.

The discriminative performance of oJSW (AUC 0.86–0.97) is comparable to the reproducibility of expert OARSJ JSN grading (intra-rater $\kappa \approx 0.86$; inter-rater $\kappa \approx 0.73$) [7]. Thus, even the lower bound of oJSW performance approximates the reliability limit of manual assessment. While an AUC of 0.86 may seem modest, it matches human-level discrimination and achieves this automatically and reproducibly, eliminating observer variability and enhancing standardisation across datasets.

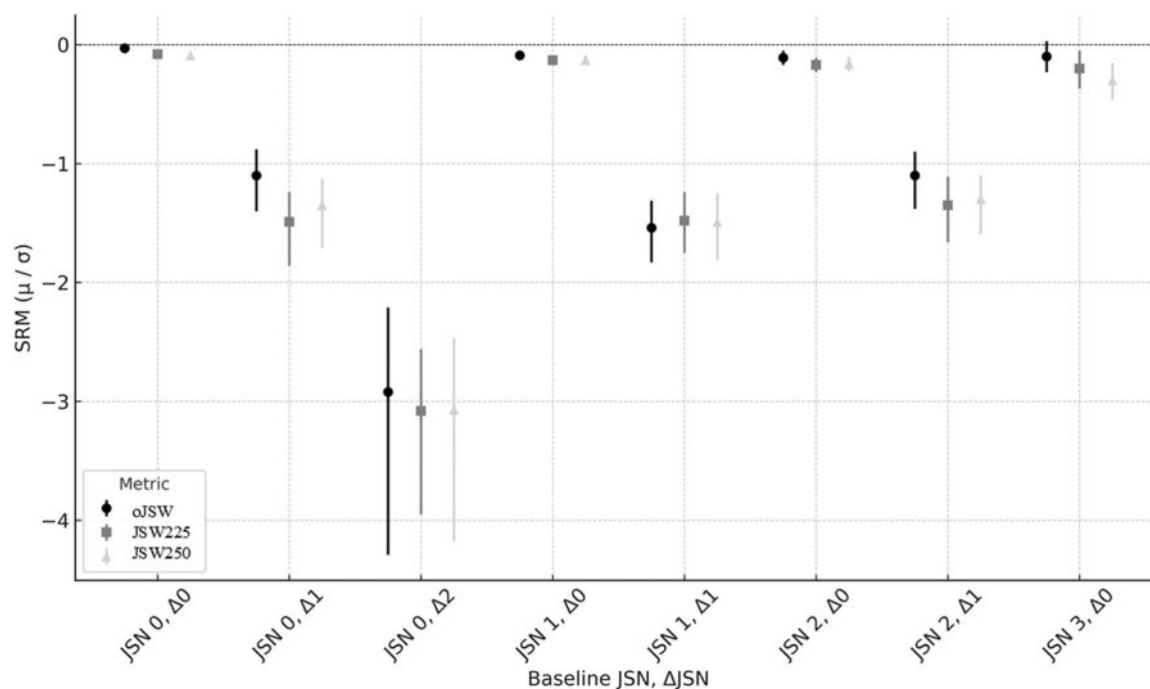


FIGURE 4 Bootstrapped standardised response means (SRMs) at 12-month intervals for oJSW, JSW225 and JSW250, stratified by baseline medial JSN grade and Δ JSN category. Points represent SRM estimates, and error bars indicate 95% confidence intervals derived from 1000 bootstrap iterations. oJSW is shown as black circles, JSW225 as grey squares and JSW250 as light grey triangles.

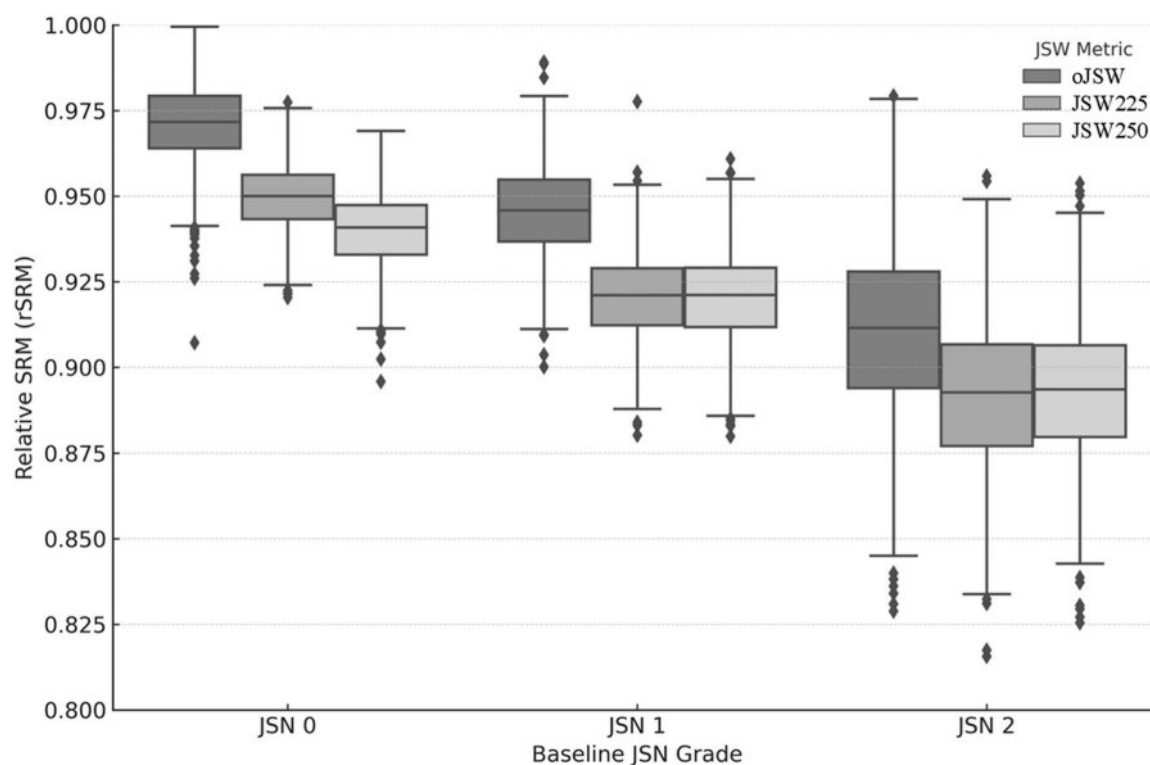


FIGURE 5 Bootstrapped relative standardised response mean (rSRM) at 12-month intervals for oJSW, JSW225 and JSW250, stratified by baseline medial JSN grades 0, 1 and 2. Box-and-whisker plots represent the distribution of rSRM estimates obtained through 1000 bootstrap resamples per subgroup.

One limitation of minimum JSW in advanced OA is the floor effect – once the joint space reaches zero (bone-on-bone contact), further narrowing cannot be detected even if cartilage loss continues elsewhere. In this study, oJSW frequently reached zero in KL grade 4 knees, limiting its ability to capture additional progression. Fixed-location measures, positioned slightly lateral to the contact point, continued to show measurable change, yielding marginally higher absolute SRMs in advanced cases but also greater variability in stable knees, indicating higher noise. The rSRM highlighted this distinction: oJSW achieved values of 0.97, 0.95 and 0.91 for baseline JSN grades 0, 1 and 2, respectively, exceeding JSW225 (0.95, 0.92 and 0.89) and JSW250 (0.94, 0.92 and 0.89). This consistently higher rSRM supports superior signal-to-noise performance across OA severities. These results extend prior work [18], favouring fixed-location JSW over traditional mJSW in moderate OA, where central cartilage remains. By adaptively identifying the narrowest point within a defined medial region, oJSW combines the flexibility of mJSW with the reproducibility of fixed-location methods. Compared with the standardised JSW (stdJSW) proposed by Paixao et al. [20], which showed slightly lower rSRM in advanced disease (~0.87 at JSN grade 2), oJSW provides greater specificity for detecting meaningful progression in severe OA.

The implications for research and clinical trials are significant. Structural endpoints remain central to the evaluation of disease-modifying OA drugs (DMOADs). Biomarkers with higher sensitivity and specificity can increase the statistical power of studies and reduce required sample sizes or study durations [21]. Particularly in early-phase or short-duration trials, oJSW may be more effective than conventional JSW metrics in detecting subtle, clinically meaningful differences in structural outcomes. The automated nature of oJSW facilitates retrospective application to existing datasets and integration into prospective workflows without changes to imaging protocols. Furthermore, clinically, oJSW offers utility for routine OA monitoring. A rapid decline in oJSW could serve as an early indicator of disease acceleration, potentially prompting timely interventions such as weight reduction, exercise modulation, or surgical consultation. Importantly, the automation and reproducibility of oJSW allow consistent, objective quantification across sites, reducing inter-reader variability and facilitating uniform assessment in multi-centre studies and electronic health record-based monitoring systems [11, 12]. This aligns with broader trends in AI-assisted musculoskeletal imaging and digital health systems.

Several limitations should be acknowledged. First, this study used data from the OAI, a well-curated but specific cohort, which may limit generalisability. All radiographs were obtained using a standardised fixed-

flexion protocol; thus, applicability to other imaging techniques, such as standing anteroposterior views, remains uncertain. External validation across diverse populations and imaging settings is warranted. Second, although oJSW provides an automated and reproducible measure of structural change, its relationship with pain, function, or arthroplasty risk was not assessed. Structural and symptomatic progression are related but variably concordant – radiographic severity correlates with pain across populations [26], yet overall associations remain modest; [2, 5] rapid deterioration before arthroplasty can coincide with symptomatic decline [22]. Third, this study focused on the medial tibiofemoral compartment, which is most commonly affected in knee OA. In advanced disease (KL grade 4), oJSW may approach zero, limiting detection of further progression; in such cases, integration with additional imaging biomarkers – such as subchondral bone curvature, contact area, or other AI-derived features (e.g., bone shape, trabecular texture, or coronal alignment) [3, 24] – may improve disease characterisation. Finally, although oJSW serves as a surrogate for cartilage loss, its biological specificity remains to be validated through comparisons with MRI-derived cartilage thickness or meniscal extrusion. Future work should examine whether accelerated oJSW decline predicts clinically meaningful outcomes.

In conclusion, oJSW represents a methodological advance that enables automated, standardised quantification of medial joint space width in knee OA. By identifying the narrowest point within a fixed anatomical zone, oJSW improves measurement precision and responsiveness relative to conventional fixed-location methods. While these findings highlight its potential utility as a structural endpoint in longitudinal studies, broader external validation and clinical correlation are needed before translation into routine clinical practice.

AUTHOR CONTRIBUTIONS

Do Weon Lee: Collection and assembly of data, statistical analysis and draft of the manuscript. **Dae Seok Song, Yong Seuk Lee and Ja-Young Choi:** Statistical analysis, writing and editing. **Du Hyun Ro:** Conception of idea, statistical analysis and editing of the manuscript. All authors revised and approved the final manuscript for submission.

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CONFLICT OF INTEREST STATEMENT

The corresponding author, Du Hyun Ro, is the CEO of CONNECTEVE Co., Ltd. However, this had no effect

on the results of our research. No other author reports conflict of interest.

DATA AVAILABILITY STATEMENT

The datasets used in the study are public-use datasets. The details about acquisition of these dataset are provided in <https://nda.nih.gov/oai/query-download>.

ETHICS STATEMENT

This article was prepared using the Osteoarthritis Initiative (OAI) public-use dataset. Ethical approval for collecting all subject information was provided by the OAI, and informed consent was obtained from all individual participants included in the study. This article does not contain any studies with human participants performed by any of the authors. All participants provided informed consent before participating, in accordance with the Helsinki Declaration. Access, download and analyses of the data were performed under the Data Use Agreement of OAI.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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